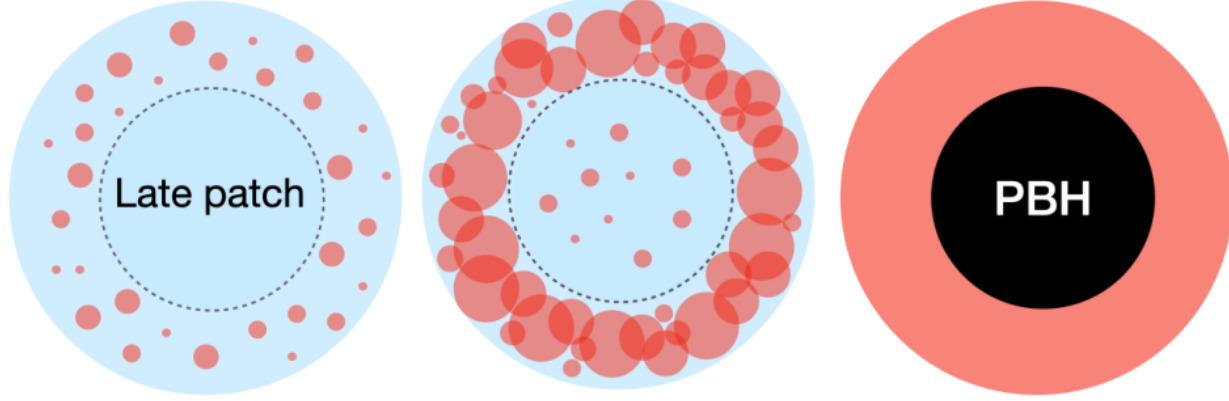


Numerical Simulations of PBH Formation via Delayed First-order Phase Transitions

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Introduction

- Randomness of nucleation (some patches nucleate late):
 $\rho_r \propto a^{-4}$, while $\rho_V = \text{const} \implies$ local overdensity



- Formation criterion: $\delta(t_H) > \delta_c \sim 0.45$ **or** $2M/R > 1$
- Need to be verified by **numerical simulations!**

Model and simulation setup

Coupled gravity-scalar-fluid system:

- $S = \int d^4x \sqrt{-g} \left[\frac{(4)R}{16\pi} - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) + \mathcal{L}_{\text{fluid}} \right]$
- $ds^2 = -A(t, r)^2 dt^2 + B(t, r)^2 dr^2 + R(t, r)^2 d\Omega^2$, with $A = 1$
- $V(\phi) = \left[1 + \lambda \left(\frac{\phi}{\phi_T} \right)^2 - (2\lambda + 4) \left(\frac{\phi}{\phi_T} \right)^3 + (\lambda + 3) \left(\frac{\phi}{\phi_T} \right)^4 \right] V_F$

Initial conditions: a **false vacuum bubble** embedded in a radiation-dominated background ($\omega = 1/3$)

$$\phi(t_i, r) = \frac{\phi_T}{2} \left[1 + \tanh \left(\frac{r - r_i}{l} \right) \right], \text{ with } l = \frac{r_i}{10} \quad (1)$$

Misner-Sharp mass and expansions:

$$M = \frac{R}{2}(1 + U^2 - \Gamma^2), \quad \Theta^\pm = \frac{\sqrt{2}}{R}(U \pm \Gamma) \quad (2)$$

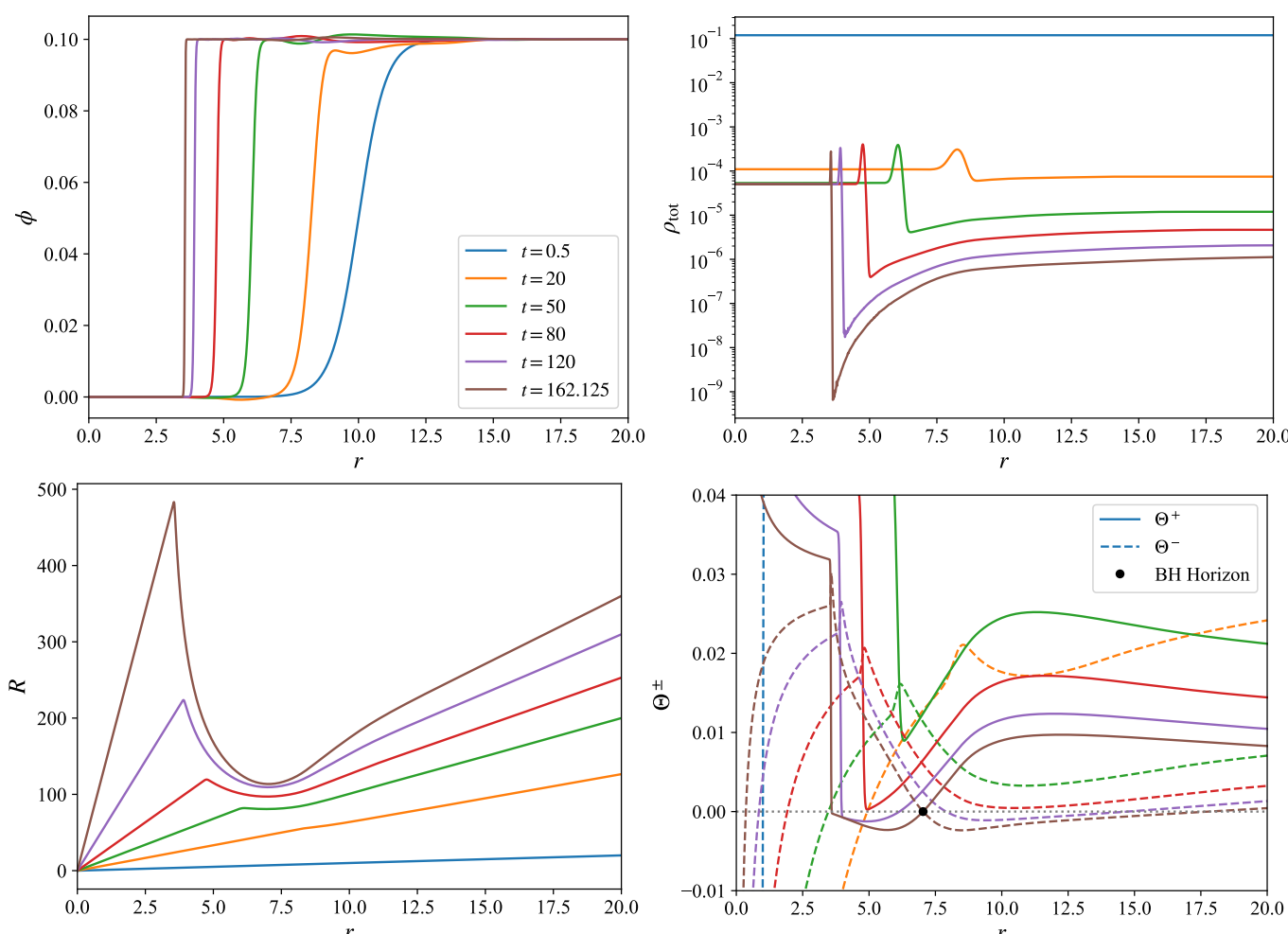
Numerical implementation:

- Finite difference method + Fourth-order Runge-Kutta method
- Adaptive mesh refinement (AMR)** to resolve the bubble wall

Dynamics of false vacuum bubbles

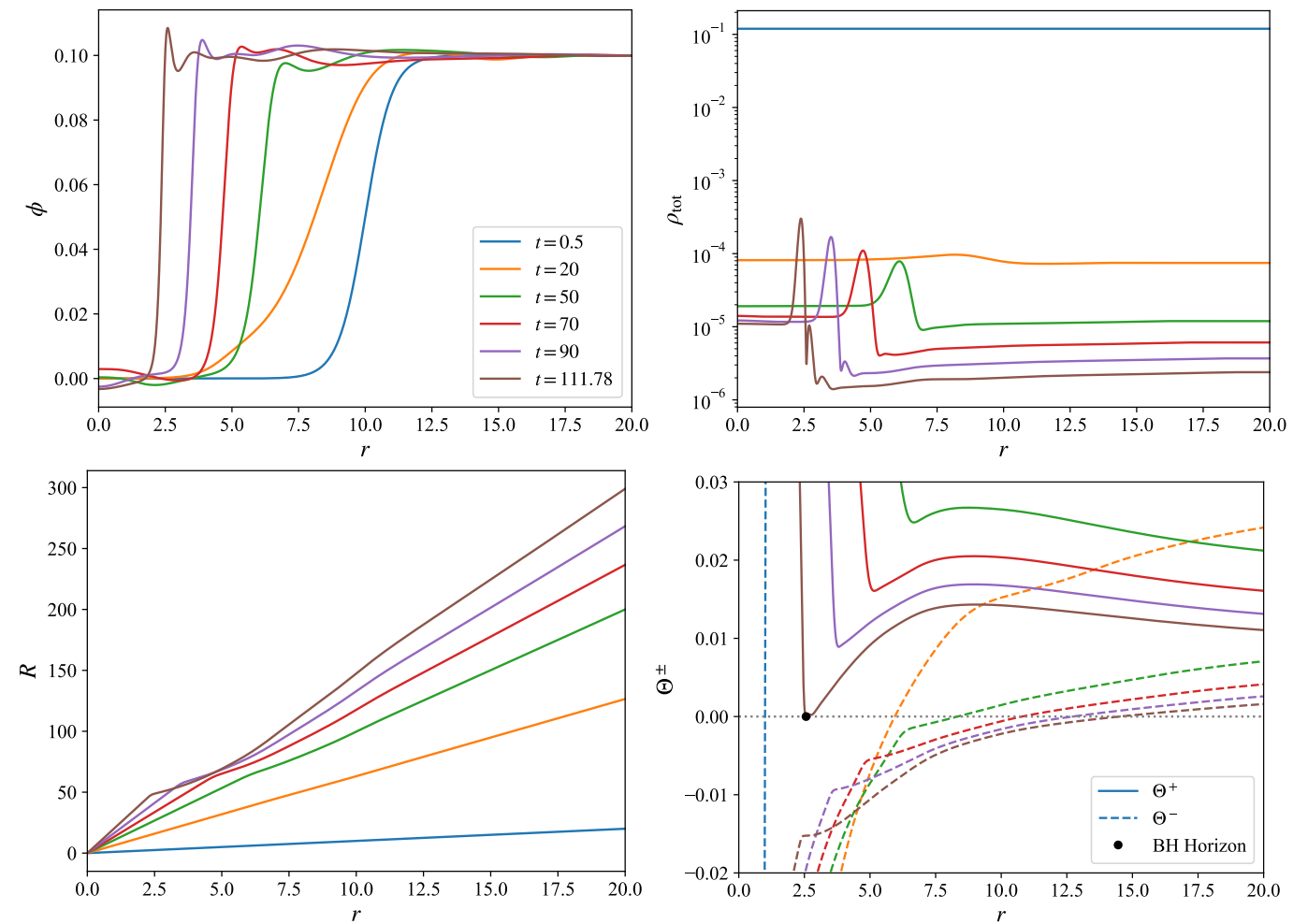
We fix $\phi_T = 0.1$, $\lambda = 20$, $r_i = 10$, and vary V_F :

- Type B (supercritical) PBH formation:** $V_F = 5 \times 10^{-5}$



Baby universe inside the PBH and bifurcating trapping horizon ($\Theta^+ = \Theta^- = 0$) are formed due to the vacuum energy

- Type A (subcritical) PBH formation:** $V_F = 1 \times 10^{-5}$



Normal PBH ($\Theta^+ = 0$ and $\Theta^- < 0$) is formed due to the bubble wall energy

- No PBH formation:** $V_F = 5 \times 10^{-7}$
FOPT completes and scalar field oscillates near the true vacuum

Thresholds for type A/B PBHs

Schwarzschild criterion is equivalent to apparent horizon:

$$\Theta^+ \Theta^- = \frac{2}{R^2}(U^2 - \Gamma^2) = 0 \implies \frac{2M}{R} = 1 \quad (3)$$

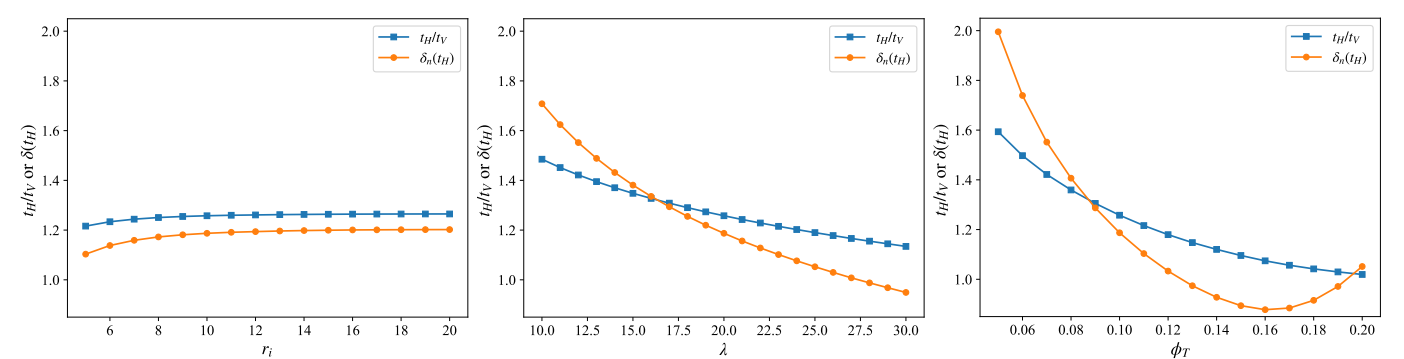
Time scale ratio t_H/t_V :

$$t_H = \frac{r_i^2}{2}, \quad t_V = \frac{1}{2} \sqrt{\frac{3}{8\pi\rho_V}} \quad (4)$$

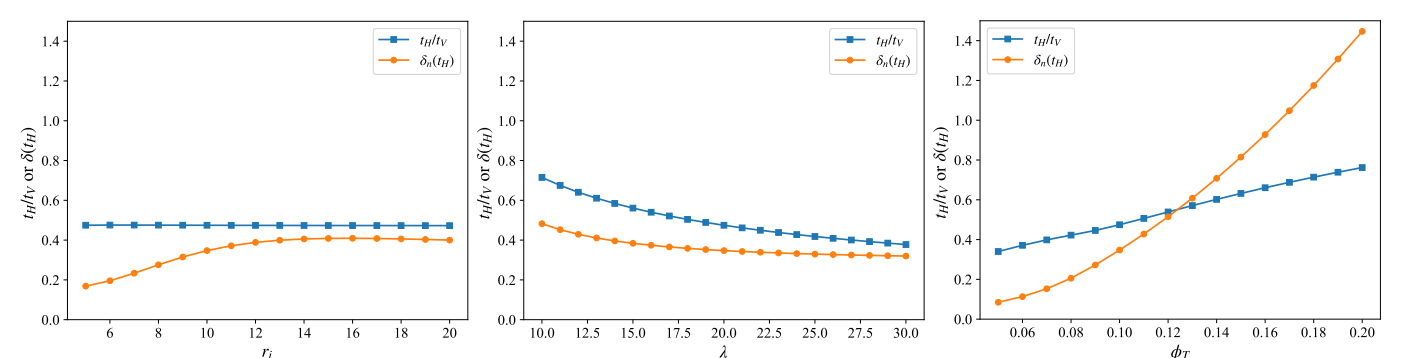
Local density contrast at the horizon crossing time:

$$\delta(t_H) = \frac{\rho_{\text{in}}(t_H) - \rho_{\text{out}}(t_H)}{\rho_{\text{out}}(t_H)} = \frac{\rho(t_H, 0) - \rho(t_H, r_{\text{max}})}{\rho(t_H, r_{\text{max}})} \quad (5)$$

- Threshold for type B PBH**



- Threshold for type A PBH**



Conclusion

- Both type A and type B PBHs can form from the collapse of false vacuum bubbles via delayed FOPTs
- The time scale ratio t_H/t_V is a more robust criterion for PBH formation than the local density contrast $\delta(t_H)$